Cross-disciplinary applications of the Toulbar2 solver

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Abstract / Introduction

Combinatorial problem solving

Cluster editing for graphs

Zahn's decision problem can be described as the cluster editing problem. Given a graph, what is the minimal number of operations (adding edges, removing edges) needed to create a disjoint union of cliques. Below an exemple of the cluster editing problem:

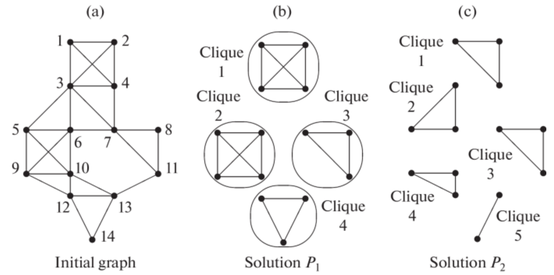


Fig. 1: Example of SMC: Robust Supply Chain with Different Network Size.

By transforming the cluster editing problem into a VCSP and solving it using Toulbar2, Mhamdi and Naanaa (2014) obtained results much better than the fixed parameter algorithm for both sparse and dense graphs.

A complicated chess problem

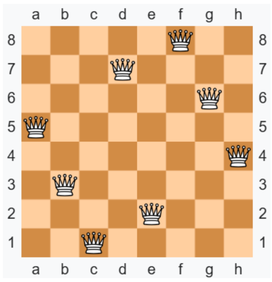


Figure 1: 8-Queens example

The N-Queen problem is a non-trivial mathematical problem that looks for a placement of N queens on a chessboard of size N by N where no two queens are threatening each other. The figure to the left shows an example of a solution where $n = 8$.

The weighted N-Queen problem is the difficult optimization problem of minimizing the cost of the N-Queen if every tile incurs a cost. By modeling the weighted N-Queen problem as a VCSP, Sewa et al. (2022) demonstrated that Toulbar2 significantly outperforms other state-of-the-art solvers.

Other problems such as the quadratic assignement problem, the knapsack problem with conflict graph and discrete integrations were also solved using the VCSP framework and showed good performance of the Toulbar2 solver.

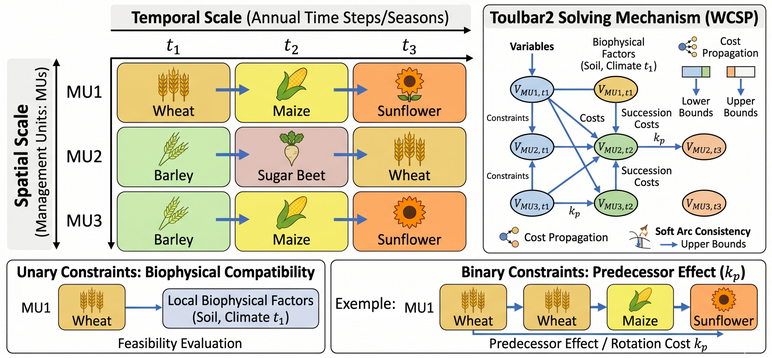
Bioinformatics & Life Sciences

marco

Recherche Opérationnelle & Planification

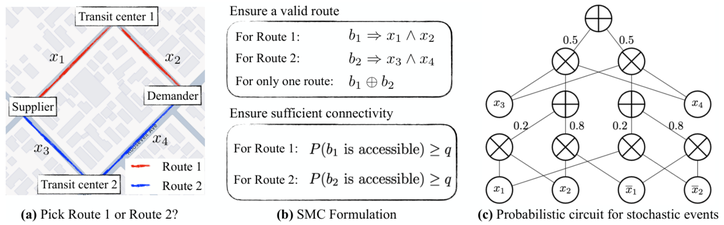
Le formalisme $WCSP$ performe dans la modélisation d'environnements complexes en combinant des règles strictes (contraintes dures) et l'optimisation de critères de qualité (contraintes souples).

Figure 1: Spatio-temporal Agricultural Modeling



Artificial Intelligence & Machine Learning

Toulbar2 struggles with real world applications since exhaustive search creates computational bottlenecks on large instances. We can integrate machine learning, introducing new ways to improve the solution quality. For example, algorithms like KOCO SMC handle large instances by tracking probability bounds during the inference process. This early evaluation prevents the solver from performing exhaustive enumerations.



(a) Pick Route 1 or Route 2? **(b)** SMC Formulation **(c)** Probabilistic circuit for stochastic events

Fig. 2: Example of SMC: Robust Supply Chain with Different Network Size.

Deep learning models can also transform raw sensory data directly into Graphical Models. This integration merges the perception of neural networks with the exact reasoning of constraint solvers to increase overall flexibility. For example, a neural network can translate unmapped protein structures or raw images into cost matrices, allowing Toulbar2 to focus exclusively on executing the exact discrete optimization.

Conclusion

Brief summary of the main takeaway from your literature review. What is the logical next step for your introductory research project?

References

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